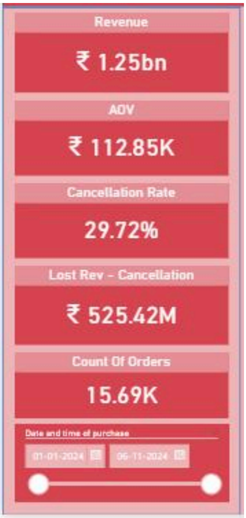
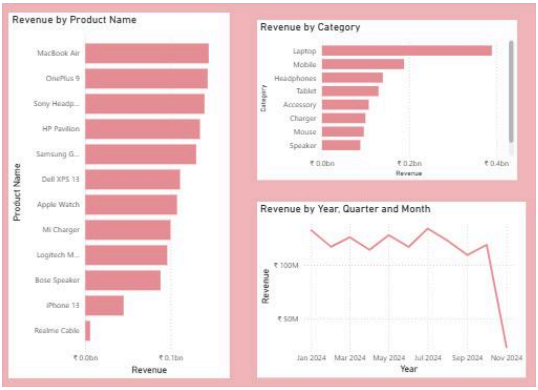
E-Commerce Sales Dashboard





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E-Commerce Sales Dashboard – End-to-End Power BI Project

Objective

Build a professional interactive e-commerce sales dashboard using Microsoft Power BI to analyze and visualize key business performance metrics — replicating a real-world business intelligence scenario where stakeholders need insights into sales performance, trends, and customer behavior.

Tools Used

- Microsoft Power BI Desktop
- Power BI Power Query Editor (for data cleaning & transformation)
- DAX (Data Analysis Expressions) for measures and calculations
- Data modeling (relationships between tables)

Core Components of the Dashboard

In the final dashboard includes these sections (typical for an e-commerce analytics project):

🔹 Key Performance Indicators (KPIs)
  - Total Sales
  - Total Profit
  - Net Orders / Quantity Sold
  - Sales growth rates
  - Average Order Value (if included)

All of these are represented visually using cards and summary visuals.

🔹 Trend Analysis
Trend visuals (e.g., line or area charts) to show:
- Monthly / yearly sales trends
- Profit trends over time

This is useful to reveal patterns and seasonality.

🔹 Category & Segment Breakdown
Interactive visualizations to show performance by:
- Product categories
- Customer segments
- Payment methods
- Shipping types

These insights help analyze which areas are performing best.

🔷 Geographical Insights
Map visuals (if included) showing how sales perform across regions (countries / states), helping stakeholders understand geographic performance.

🔷 Filters & Interactivity
Dashboard elements (slicers and filters) enable users to interactively slice data by date range, category, or other dimensions.

Technical Skills Demonstrated

Data Import & Transformation

Imported datasets and cleaned them using Power Query (handling missing values, formatting dates, data types, etc.).

Data Modeling

Created relationships between multiple tables (e.g., Orders, Customers, Products).

DAX Measures

Used DAX to calculate dynamic metrics such as:
- Total Sales
- Year-to-Date (YTD) sales
- Profit margins
- Quantity measures

(DAX is essential for creating actionable insights).

Visualization Techniques

Used a variety of charts and interactive visual elements like:
- Bar/column charts
- Line charts
- Map visuals
- Card visuals

to make complex data intuitive and easy to explore.

Business Insights

✔ Which category contributes the most to total sales
✔ Which payment mode is most common
✔ Seasonal performance trends (e.g., peak months)
✔ Regional performance differences

Project Summary

Designed an interactive ecommerce dashboard using Power BI to analyze sales performance and customer behavior. Leveraged data cleaning using Power Query, built a relational data model, created dynamic DAX measures, and visualized key business metrics such as total sales, profit trends, category performance, and geographic insights. The dashboard helps stakeholders monitor performance, compare segments, and make data-driven decisions in real time.